Himalaya Carbon Tokenization Engine

A plain-language research note on viability, uniqueness, and implementation in Bhutan

1. What this project is really about

The **Himalaya Carbon Tokenization Engine** should not be described as "another carbon registry" or "another NFT carbon platform." That would be a strategic own-goal.

The strongest version of the project is this:

Bhutan's official registry and Article 6 systems stay as the source of truth.
Himalaya becomes the market layer that makes those official carbon units easier to trade, track, and retire.

That distinction matters a lot.

Bhutan already has unusually strong foundations for carbon-market participation. Its Constitution requires at least **60% forest cover for all time**, and Bhutan's official reporting shows that in 2020 it emitted about **2,723 Gg CO2e** while maintaining sink capacity of about **9,513 Gg CO2e**, leaving it net negative. Bhutan also launched its **National Carbon Registry** at COP28 and became the **first national registry** to fully integrate with **CAD Trust**. On top of that, Bhutan and Singapore signed an **Article 6 Implementation Agreement** on **28 February 2025**, and project applications under that route opened from **1 December 2025**. (Department of Local Governance)

So the country already has the legal forest base, the climate accounting base, the registry base, and an international transfer pathway. That is not small. That is the hard plumbing.

The open question is not whether Bhutan can support digital carbon infrastructure. It already can. The real question is whether **Himalaya adds a new layer of value** without creating a parallel system that turns the whole thing into carbon-flavored spaghetti. (Climate Action Data Trust)

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2. The simple conclusion first

Feasibility verdict

Yes, the Himalaya project can work.
But only if it is designed as:

- a **secondary-market and retirement layer**, not a rival issuance platform;
- a **registry-synchronized token system**, not a synthetic carbon coin factory;
- a **government-recognized bridge** between official registry units and public-chain trading tools;
- and a **high-integrity Article 6-compatible infrastructure**, not generic web3 carbon theater. (Carbon Markets Cooperation)

That is the sane architecture.

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3. Why Bhutan is a strong base for this idea

Bhutan is unusually well positioned because its carbon story is not just branding. It sits on law, inventory data, and state infrastructure.

The constitutional part is famous for a reason: Bhutan is legally required to maintain at least **60% forest cover**. That gives long-term credibility to its forest sink story. Official UNFCCC reporting then adds the quantitative side: Bhutan's 2020 gross emissions were around **2,723 Gg CO2e**, while sink capacity was around **9,513 Gg CO2e**, making the country net negative that year. (Department of Local Governance)

Then comes the infrastructure part. Bhutan launched its **National Carbon Registry** at COP28 and integrated that registry with **CAD Trust**, which is designed to harmonize registry data and reduce double-counting risk. CAD Trust describes itself as a decentralized metadata platform built to improve transparent accounting in line with Article 6. (Climate Action Data Trust)

Then comes the market access part. The **Singapore–Bhutan Article 6 Implementation Agreement** created a bilateral route for authorization, issuance, transfer of ITMOs, and corresponding adjustments. Singapore's government has also said that authorized Bhutan credits can be used by Singapore carbon-tax-liable companies, within the limits of Singapore's International Carbon Credit framework. That gives Himalaya something precious: an actual demand channel, not just vibes and pitch-deck fog. (Carbon Markets Cooperation)

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4. What problem Himalaya should solve

The global voluntary carbon market has real problems. Not imaginary consultant problems—actual market structure problems.

CFA Institute notes that the VCM is held back by **mistrust in credit quality**, **lack of standardization**, **lack of transparency**, and the fact that roughly **two-thirds of transactions are private**. The World Bank's 2025 carbon-pricing update also says carbon credit supply continued to outstrip demand, pushing the pool of unretired credits to **almost 1 billion tons in 2024**. That means this is not a clean, elegant market. It is patchy, opaque, and full of credibility disputes.

That is where Himalaya has a valid opening.

The project should target four specific market pain points:

1. Fragmentation
Carbon data, credit status, and trading are often spread across registries, brokers, and private contracts. (CFA Institute Research and Policy Center)

2. Poor transparency
Price discovery is weak because so much trading is private.

3. Liquidity problems
Even good credits can be hard to buy in the exact quantity a user wants. (CFA Institute Research and Policy Center)

4. Double-counting and lifecycle confusion
This is the dragon in the basement. If tokenization does not follow registry rules, credibility collapses fast. Verra explicitly prohibited tokens based on **retired credits**, and Gold Standard has required **express written consent** and formal conditions for tokenization because uncontrolled tokenization raises integrity, security, and regulatory risks. (Verra)

So the Himalaya opportunity is real—but only in a design that respects registries first and blockchain second.

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5. What Bhutan already has, and why that matters

Bhutan is not starting from zero.

Public reporting says **Druk Holding & Investments (DHI)** partnered with **InfraBlocks Technologies** to build a blockchain-based carbon credit platform, with an MVP reportedly developed with **ADB** support and backed by a **MAS proof-of-concept grant**. Public descriptions of that initiative frame it around streamlining issuance, recording issuer/verifier data, and improving authenticity assurance for carbon credits. Because this is mainly described in public reporting and company-linked press material rather than a full technical whitepaper, it should be treated carefully—but the directional point is clear: Bhutan has already experimented with blockchain-backed carbon issuance infrastructure. (EIN Presswire)

This is exactly why Himalaya must not pitch itself as "Bhutan's blockchain carbon platform."

That role is already occupied, at least in concept and partly in execution.

The clean positioning is:

- **existing state-linked infrastructure = issuance / registry digitization / primary workflow**
- **Himalaya = secondary trading / fractional exposure / retirement UX / public-chain market layer**

That is synergy, not duplication. And duplication is the thing that would kill the project politically and architecturally.

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6. The biggest design mistake to avoid

Here is the key technical truth:

CAD Trust is not the legal registry of record.

It is a **metadata layer**, not the legal engine that creates, retires, or transfers the underlying mitigation outcome. CAD Trust's own description is about linking, aggregating, and harmonizing registry data to improve transparency and avoid double counting. Its new **Data Model v2.0 API** is for sharing and accessing harmonized data, not for magically minting legal carbon units out of the digital mist. (P3 Resource Center)

So the Himalaya engine **cannot safely mint "from CAD Trust."**

It must mint only from **registry-held units** that are:

- officially issued,
- uniquely identified,
- not already transferred elsewhere,
- not already retired or cancelled,

- and, where relevant, properly authorized for Article 6 transfer. (Carbon Markets Cooperation)

This is the central architectural rule. Ignore it, and the whole project becomes a fancy counterfeit machine with a dashboard.

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7. The best architecture in easy words

Here is the most defensible version of the system.

Layer 1: The official registry stays in charge

Bhutan's national registry remains the legal system of record for carbon units. It tracks the real unit status: issued, authorized, first transferred, cancelled, retired, and used. The Singapore–Bhutan Implementation Agreement also requires registry handling of unique identifiers, vintage, transfer history, and voluntary cancellation. (Carbon Markets Cooperation)

Layer 2: CAD Trust is the public verification spine

CAD Trust gives a standardized metadata view that outside users, developers, and auditors can check. Its API v2.0 now exists for public access and testing, which makes automated integration far more realistic than before. (Climate Action Data Trust)

Layer 3: Himalaya becomes the tokenization and trading layer

Himalaya should create an on-chain token only when a corresponding registry unit is **immobilized** or locked in a way that prevents double use elsewhere. In plain English: if one carbon unit is represented on chain, that same unit cannot remain freely tradable off chain at the same time.

Layer 4: Retirement must happen in both worlds

When a user "retires" a token:

1. the on-chain token is burned or locked;
2. the registry records the matching cancellation/retirement;
3. CAD Trust and the dashboard reflect the updated status;
4. the user gets an auditable proof trail. (Carbon Markets Cooperation)

That is the correct two-ledger model.

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8. ERC-1155 or ERC-20? The honest answer is: both

Your original concept leans toward **ERC-1155**, and that part is sensible.

ERC-1155 is good because a single contract can manage many token classes. That fits carbon well, because each project/vintage/methodology bucket can have its own token ID. One token ID could mean something like:

Bhutan Forestry Project X / 2025 vintage / Article 6 authorized

That preserves useful metadata while still allowing units inside that bucket to behave in a fungible way.

But there is an important catch: most mainstream AMM DEX systems are built around **ERC-20** assets. Uniswap's documentation and pair contracts are ERC-20-based. So if Himalaya wants smooth DeFi pool integration, pure ERC-1155 on its own is awkward. (Uniswap Docs)

So the stronger architecture is:

- **ERC-1155 as the canonical registry-aware token layer**
- **ERC-20 wrappers for trading pools**

That gives you the best of both worlds:

- rich carbon metadata on the canonical side,
- easy AMM trading on the liquidity side.

This is much better than pretending ERC-1155 alone solves everything. The blockchain goblin always hides in the interface layer.

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9. What fractionalization should really mean

The original concept talks about fractional carbon purchases. That is good in user-experience terms, but you have to be honest about registry reality.

Many registries work in **whole units**, usually whole tons. If that is the case, then "fractional retirement" is not automatically a registry-native action.

So the safer first version is:

- allow fractional **trading exposure** on-chain,
- but only allow final **redemption/retirement** in whole units,
- or use a managed batching process where many smaller users are combined into one whole-unit retirement event.

This sounds less sexy than "buy 0.37 tCO_2 and retire instantly," but it is much more likely to survive contact with legal and registry rules. And surviving contact with reality is underrated.

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10. What the oracle should actually read

Another common mistake is to think the oracle should read "the forest" directly and then mint credits from live sensor data.

That is not how high-integrity carbon accounting works.

For tradable units, the real source of truth is not raw satellite data. It is the **official registry state** after verification and authorization.

So the core oracle should primarily pull or attest to:

- project ID
- unit or batch ID

- vintage
- authorization status
- active / immobilized / transferred / cancelled / retired status
- whether corresponding adjustment applies on first transfer, where relevant. (Carbon Markets Cooperation)

MRV data from forests, satellites, sensors, or national systems is still useful—but mainly as a **monitoring and risk flag layer**, not the direct legal trigger for minting. In other words:

- **registry data decides unit state**
- **MRV data informs risk, alerts, and possible pauses**

That is a much stronger model.

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11. How Bhutan's MRV systems fit in

Bhutan's climate policy and reporting framework already emphasize MRV systems and monitoring of sink performance. The Climate Change Policy calls for project-level MRV systems, and Bhutan's broader carbon-market development work has included registry and monitoring capacity building. World Bank commentary on Bhutan's carbon-market development also points toward improved use of satellite, drone, and sensor data to strengthen MRV. (FAOLEX)

So for Himalaya, the practical role of MRV is:

- detect anomalies,
- flag project degradation risks,
- improve investor confidence,
- support faster verification cycles,
- and help decide whether a token class should be paused pending review.

That is valuable. But it should not replace the registry as the legal status engine.

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12. Why the project can still be unique even though Bhutan already tokenizes

This is the user's core question, and it deserves a sharp answer.

"Will this work if Bhutan already has a blockchain carbon initiative?"

Yes—if Himalaya is the market infrastructure around official units, not a separate source of units.

The uniqueness does **not** come from "we tokenize carbon." A lot of people have done that already, with mixed results and occasional clown-car outcomes.

The uniqueness comes from three deeper things:

A. Sovereign-grade provenance

Minting is tied to Bhutan's registry state and, where relevant, Article 6 authorization—not just to a private bridging process. (Carbon Markets Cooperation)

B. Public metadata transparency

Each on-chain asset points back to CAD Trust-linked metadata, improving auditability and independent verification. (Climate Action Data Trust)

C. Accounting-aware lifecycle

The token mirrors meaningful legal states: authorized or not, first transferred or not, retired or not, cancelled or not. That is much more serious than a generic "carbon NFT." (Carbon Markets Cooperation)

So the real novelty is not the token standard.
The real novelty is making the token a **faithful digital representation of a sovereign registry unit**.

That is the thing.

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13. Recommended product shape for the MVP

A practical MVP should be modest and disciplined. Not every hackathon dream needs to become a sentient carbon bazooka on day one.

Best MVP components

1. Permissioned minting module
Only a recognized operator or government-approved gateway can mint tokenized units. This is essential for integrity.

2. Registry-sync service
Continuously checks official status before minting, transfer-sensitive actions, and retirement.

3. CAD Trust metadata integration
Displays project details, vintage, registry references, and status proofs using standardized data. (Climate Action Data Trust)

4. ERC-1155 canonical asset layer
Each token ID represents a specific credit class.

5. ERC-20 liquidity wrappers
Optional wrapper contracts for DEX trading.

6. Corporate retirement dashboard
Shows purchase, burn, registry cancellation, and public proof trail.

7. Compliance-first admin controls
Pause, blacklist, recall, or dispute-handling functions for exceptional cases such as authorization revocation or verification failure. The Singapore–Bhutan agreement explicitly contemplates revocation and dispute handling, so the smart-contract system should not pretend reversals are impossible. (Carbon Markets Cooperation)

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14. Regulatory fit in Bhutan

Bhutan has already set rules for carbon-market participation through its **Carbon Market Rules 2023**. Those rules define an **Approved GHG Program** as the infrastructure for listing, registering, issuing, holding, transferring, and retiring mitigation outcomes, and they define the **Letter of Authorization** pathway for participation under cooperative approaches. They also identify corresponding adjustment as the mechanism to prevent double counting.

That means Himalaya cannot just show up in a hoodie and declare itself legitimate because it has Solidity code and a nice logo.

It would need some combination of:

- recognition or acceptance within the official Bhutan carbon-market framework,
- operational alignment with DECC / MoENR processes,
- and strong controls for KYC/AML if trading and retirement involve corporate or regulated entities.

Bhutan also has a **FinTech Regulatory Sandbox** under the RMA, and the IMF published a review of that framework in February 2026. The sandbox exists to test financial innovation in a controlled way, but the IMF review noted governance, capacity, and oversight challenges in the current framework. So a Himalaya proposal would need to present itself as something that adds genuine new functionality, not just a duplicate of existing tokenization efforts. (RMA)

So yes, there is a regulatory path—but it is a path for a **controlled, permissioned, integrity-heavy system**, not a fully anarchic DeFi carnival.

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15. Commercial reality check

Here is where the original draft needed some pruning.

The **$10.5 trillion by 2034** figure is the kind of number that looks glorious in a pitch deck and immediately makes careful readers narrow their eyes. Mainstream research tends to be much more conservative for the VCM specifically.

CFA Institute, citing Morgan Stanley, references projections from about **$2 billion in 2022** to roughly **$100 billion by 2030** and around **$250 billion by 2050** for the VCM. That is still huge. It just is not galaxy-brain huge. For practical planning, those are safer anchors than ultra-bullish trillion-scale assumptions.

The more grounded business case for Himalaya is not “the whole world will trade on our DEX.”

It is this:

- Bhutan has a rare high-integrity supply story,
- Article 6 gives it a serious policy route,
- Singapore provides an early demand wedge,
- and there is room for a better trading + retirement + audit UX around official units. (Carbon Markets Cooperation)

That is credible.

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16. Monetization model that makes sense

A sane revenue model would have three pieces:

1. Minting / bridge fees

Charged when official registry units are converted into on-chain tradable form.

2. Trading fees

Charged on swaps or liquidity pool activity.

3. Retirement / enterprise dashboard fees

Charged for corporate-grade retirement certificates, audit views, API usage, reporting, or treasury integrations.

This is much better than relying only on speculative token appreciation, which is usually where otherwise promising climate-tech projects wander into the swamp.

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17. Biggest risks

The biggest risks are not mainly coding risks.

Governance risk

If government recognition is unclear, the token loses legitimacy fast.

Registry conflict risk

If token supply ever exceeds immobilized registry units, the project is in deep trouble.

Standard-setter risk

Verra and Gold Standard have both shown that tokenization without controlled rules can be restricted, prohibited, or consent-gated. (Verra)

Claims risk

Even with perfect unit tracking, corporate claims depend on how authorization and corresponding adjustments are handled. Himalaya must expose those statuses clearly, not bury them in fine print. (Carbon Markets Cooperation)

Liquidity risk

A DEX is only useful if there are enough buyers, sellers, and trusted market makers. Otherwise it is just a decorative pond.

Oracle trust risk

The oracle must deliver authoritative registry facts, ideally backed by source signatures or government attestations. Otherwise the oracle becomes the hidden sovereign, which is absurd and fragile.

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18. Final recommendation

Here is the clearest strategic framing:

Himalaya should be positioned as:

Bhutan's registry-synchronized carbon market layer for trading, transparency, and retirement

Himalaya should not be positioned as:

a new carbon registry, a parallel token source, or a free-floating carbon coin platform

That one distinction changes everything.

If built the right way, Himalaya can be genuinely valuable because it would sit on top of Bhutan's strongest advantages:

- constitutional forest protection,
- net-negative national profile,
- a national carbon registry,
- CAD Trust integration,
- and an operational Article 6 route with Singapore. (Department of Local Governance)

If built the wrong way, it becomes just another carbon token experiment that collides with registry rules and loses credibility.

So the elegant version is not:
"mint carbon on chain."

It is:
"represent official Bhutan carbon units on chain, without breaking accounting, law, or trust."

That is the version that has teeth.

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19. One-line executive summary

Himalaya is viable only if it acts as Bhutan's secondary liquidity and retirement layer for officially issued, registry-backed carbon units—not as a separate carbon issuance platform.